LINEAR ANTI-RECURRENCE SEQUENCES ARE LINEAR PLUS AUTOMATIC

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ABSTRACT. Let $\mathbf{a} = (a_1, \ldots, a_k)$ be a vector of positive integers, $k \geq 2$, with $\tau = \sum a_j$. The anti-recurrence sequence of $\mathbf{a}$ is the unique increasing sequence (A_n) such that $A_n = \sum_j a_j B_{k(n-1)+j}$, where (B_m) enumerates the complement of $\{A_n\}$ in $\mathbb{N}$. The case $\mathbf{a} = (1,1)$ is the anti-Fibonacci sequence (OEIS A075326). Fokkink and Joshi conjectured that $A_n - \kappa n$ is always τ-automatic, where $\kappa = k\tau + 1$. They proved this under a restriction on $\mathbf{a}$ ("A_1-boundedness") and asked in particular about the "rusty numbers" $\mathbf{a} = (1, d)$, $d \geq 4$. We prove the conjecture for every $\mathbf{a}$; thus every such A_n is a linear sequence plus a τ-automatic sequence. More precisely, we show that $-k\tau \leq A_n - \kappa n \leq \tau - k$ for all n. We also show that $d(n) = A_n - \kappa n$ satisfies an explicit two-term recursion $d(\tau q + r + 1) = G_r(d(q), d(q+1))$, which yields a most-significant-digit-first τ-automaton with at most $1 + (k\tau + \tau - k + 1)^3$ states. For the 4-rusty numbers $\mathbf{a} = (1,4)$ we give the minimal automaton (17 states in our indexing convention, 9 in that of Fokkink and Joshi) and an independent machine-checked proof in the theorem prover Walnut. As a corollary we explain why the arithmetic progression $A_{5q+1} = 9 + 55q$ observed by Fokkink and Joshi first fails at A_{1741}. We also give Walnut proofs for $(1,5)$, $(2,7)$ and $(1,1,5)$, and report an extensive computational check of the steps of the proof on more than two thousand forms.

1. INTRODUCTION

1.1. **Anti-recurrence sequences.** The *anti-Fibonacci numbers* $0, 3, 9, 13, 18, 23, 29, \ldots$ form OEIS sequence A075326 [16]. According to the OEIS record, the sequence was entered by A. Murthy in 2002 (originally defined via the pairs of A075325); the present definition is due to P. Lallouet (2008), and the name "anti-Fibonacci numbers" to D. R. Hofstadter (2014), whose note [9] records observations about it. Each new term is the sum of the two smallest positive integers that are neither terms of the sequence nor already used in an earlier sum. Zaslavsky [23, Thm. 1] gave a closed formula: indexing so that $\bar{F}_0 = 3$, one has $\bar{F}_n = 5n+3$ for even n, while for odd n the value is $5n+3$ or $5n+4$ according to whether $(n-1)/2$ is *utterly odd*, i.e. whether the terminal run of 1's in its binary expansion has odd length.[1] Equivalently, $A_n = 5n - 2 + \mathrm{PD}_{n-1}$ for $n \geq 1$, where $\mathrm{PD} = 0, 1, 0, 0, 0, 1, 0, 1, \ldots$ is the period-doubling sequence (OEIS A096268, indexed from 0) [5, eq. (2)], [6, §1]. Bosma et al. [3, §3] verified Zaslavsky's description with Walnut. Kimberling and Moses [12] placed this example inside a general theory of systems of complementary equations, generalised it to anti-m-nacci systems, and conjectured substitutions generating the difference sequences. Bosma, Bruin, Fokkink, Grube, Reuijl and Tromp [3] proved

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[1] Read literally with the definition of "utterly odd" given in [23], the two odd cases of [23, Thm. 1] are interchanged: for example $\bar{F}_1 = 9 = 5 \cdot 1 + 4$, although 0 is not utterly odd. The correct reading, $\bar{F}_n = 5n + 4$ for odd n exactly when $(n-1)/2$ is *not* utterly odd, agrees with the sequence as listed in [23] and with the period-doubling form below; we checked both against the first 3000 terms.

the anti-tribonacci and anti-tetranacci cases with the automatic theorem prover Walnut [15]. They also formulated what they called the *Clergyman's Conjecture* for anti-k-naccis.

Fokkink and Joshi [5, 6] introduced the general notion. Let $\mathbf{a} = (a_1, \dots, a_k)$ be a positive integral linear form of dimension $k \geq 2$. Its *trace* is $\tau = \sum_j a_j$. Put $K = k\tau$ and $\kappa = K + 1$. The *anti-recurrence sequence* of $\mathbf{a}$ is the pair of complementary, strictly increasing sequences $(A_n)_{n\geq 1}$, $(B_m)_{m\geq 1}$ in $\mathbb{N} = \{1, 2, \dots\}$ such that

$$A_n = \sum_{j=1}^{k} a_j \, B_{k(n-1)+j} \qquad (n \geq 1). \tag{1}$$

Equivalently, B is generated greedily by the mex rule. Such a pair exists and is unique [5, Lemma 3], [6, Lemma 2]. The first term is $A_1 = \sum_j j\, a_j$, which we denote by A_1.

Conjecture 1.1 (Fokkink–Joshi [5, Conj. 4], [6, Conj. 2])**.** *For every positive integral linear form* $\mathbf{a}$ *of dimension* $k > 1$*, the sequence* $A_n - \kappa n$ *is* τ*-automatic.*

The two versions number this differently: it is Conjecture 4 in arXiv v1 and Conjecture 2 in the published Integers version. Both versions call the statement "$A_n - \kappa n$ is bounded" its *weak form*, and observe that Kimberling and Moses phrase their conjectures in this form. Fokkink and Joshi proved Conjecture 1.1 in the following cases:

- anti-k-bonaccis, $\mathbf{a} = (1, \dots, 1)$ [5, Thm. 8], [6, Thm. 3];
- A_1-bounded forms, $\mathsf{A}_1 \leq (k-1)\tau + 2$ [5, Thm. 12], [6, Thm. 4];
- the anti-Pell form $(1, 2)$ and the anti-Jacobsthal form $(2, 1)$, by Walnut.

For $k = 2$, A_1-boundedness means $a_2 \leq 2$. Their final section asks: "Does the conjecture hold without the restriction of A_1-boundedness? Are the rusty numbers sums of linear sequences and automatic sequences?" [6, §6].

1.2. **The rusty puzzle.** For $\mathbf{a} = (1, d)$, Fokkink and Joshi call the A_n the *d-rusty numbers*. For $d = 3$ they display a guessed 4-DFAO, i.e. an automaton reading base-4 input ($\tau = 4$), with 6 states [6, Fig. 7], and remark that their Walnut method can be applied to it. Note that $(1, 3)$ is not A_1-bounded. For $d = 4$ the sequence begins

$$9, 19, 29, 39, 54, 64, 74, 84, 98, 109, 119, 129, 139, 154, \dots$$

They observe that $A_{5n+1} = 9 + 55n$ holds "up until $n = 348$, when $A_{1741} \neq 9 + 55 \cdot 348$". They also note that the structural feature behind their $(1, 3)$ automaton, namely all 0-transitions returning to the initial state, fails for $(1, 4)$. Indeed, "proving or disproving the conjecture for the rusty numbers remains a challenge" [6, §5].

1.3. **Results.** Throughout, $d(n) := A_n - \kappa n$ for $n \geq 1$, and $d(0) := 0$ (the convention $A_0 = 0$).

Theorem A (Boundedness)**.** *For every positive integral linear form of dimension* $k \geq 2$ *and every* $n \geq 1$*,*

$$-k\tau \;\leq\; A_n - \kappa n \;\leq\; \tau - k.$$

Moreover, there is a finite list of explicit inequalities (Proposition 3.2) whose integer solutions $[L, U]$ *all satisfy* $L \leq d(n) \leq U$*. In every form we tested, the narrowest such box equals the exact range of* d *(Remark 3.5).*

Theorem B (Two-term recursion)**.** *For* $0 \leq r < \tau$ *and* $y_0, y_1 \in \mathbb{Z}$ *put*

$$G_r(y_0, y_1) = \mathsf{A}_1 - K - 1 - r - \tau + \sum_{j=1}^{k} a_j \Big([\, y_0 \leq kr + j - 1 \,] + [\, y_1 \leq kr + j - 1 - K \,] \Big),$$

where $[\,P\,] \in \{0,1\}$ *is the Iverson bracket. Then* $d(\tau q + r + 1) = G_r\big(d(q), d(q+1)\big)$ *for all* $q \geq 0$ *and* $0 \leq r < \tau$.

Theorem C (Automaticity)**.** $(d(n))_{n\geq 0}$ *is* τ*-automatic. It is generated by an explicit msd-first* τ*-DFAO with at most* $1 + (k\tau + \tau - k + 1)^3$ *states. In particular Conjecture 1.1, and hence its weak form, holds for all positive integral linear forms.*

For the 4-rusty numbers, Theorem B specialises to

(2)
$$d(5q+j) = -j-6+[\,d(q) \leq 2j-2\,]+4[\,d(q) \leq 2j-1\,]+[\,d(q+1) \leq 2j-12\,]+4[\,d(q+1) \leq 2j-11\,]$$

for $q \geq 0$ and $1 \leq j \leq 5$, and d takes values in $[-6, 1]$. In Section 4 we show:

- from (2), $A_{5q+1} = 9 + 55q$ if and only if $d(q) \neq 1$;
- the first index with $d(q) = 1$ is $q = 348$, reached by the chain $3 \to 14 \to 70 \to 348$ (Corollary 4.1);
- the minimal automaton has 17 states in our convention (Table 1), and 9 states in the convention of [6] (Section 4);
- Walnut certifies directly, without using Theorems A–C, that this automaton generates the anti-recurrence sequence of $(1, 4)$ (Theorem 4.2).

1.4. **Method.** The proof is elementary. Section 3 rests on a counting identity (Lemma 3.1): $B_m = m + \#\{i : A_i < B_m\}$, and $A_i < B_m$ holds exactly when $Ki + d(i) \leq m - 1$. This turns (1) into a self-referential formula expressing $d(n)$ through counts of earlier $d(i)$ below thresholds. An induction gives an invariant box. A short estimate shows that $[-K, \tau - k]$ is always such a box. With that box, all but two terms of each count are determined, which gives Theorem B. Theorem C follows by a standard argument: triples of consecutive values form the state of an automaton.

2. Preliminaries

We use the notation of the introduction: $\mathbf{a} = (a_1, \ldots, a_k)$ with $k \geq 2$ and $a_j \geq 1$; $\tau = \sum_j a_j \geq k \geq 2$; $K = k\tau$; $\kappa = K + 1$; $\mathsf{A}_1 = \sum_j j a_j$. Then

$$\tau \;\leq\; \mathsf{A}_1 \;\leq\; K - (k-1), \tag{3}$$

since $K - \mathsf{A}_1 = \sum_j (k-j)a_j \geq \sum_{j<k}(k-j) \geq k-1$. The first block is $B_j = j$ for $j \leq k$, hence $A_1 = \mathsf{A}_1$ and $d(1) = \mathsf{A}_1 - K - 1$. We use the gap bound $A_{n+1} - A_n \geq K$ [5, Lemma 2], [6, Lemma 1]. In fact only $A_{n+1} - A_n \geq 2$ is needed below.

A sequence $(s(n))_{n\geq 0}$ is τ*-automatic* if there is a deterministic finite automaton with output (DFAO) that, reading the base-τ representation of n, ends in a state whose output is $s(n)$. Our automata read the most significant digit first and allow leading zeros, as Walnut does. The reading direction does not affect the class of automatic sequences [1, Thm. 5.2.3]. Our automata handle leading zeros explicitly (Theorem C), so no further citation is needed for them. Equivalently, s is τ-automatic if and only if its τ*-kernel* $\{(s(\tau^e n + b))_{n\geq 0} : e \geq 0,\ 0 \leq b < \tau^e\}$ is finite [1, Thm. 6.6.2]; see also [4]. We use the kernel characterisation only in Remark 3.7.

3. The general theorem

3.1. **A counting identity.** For $m \geq 1$ let $c(m) := \#\{i \geq 1 : A_i < B_m\}$.

Lemma 3.1.

(a) $B_m = m + c(m)$. *Moreover* $A_i < B_m$ *if and only if* $Ki + d(i) \leq m - 1$.
(b) *Let* $n \geq 1$ *and* $m_j := k(n-1)+j$ *for* $1 \leq j \leq k$. *Then no index* $i \geq n$ *satisfies* $A_i < B_{m_j}$.

(c) *Write* $n-1=\tau q+r$ *with* $0\le r<\tau$, *and put* $u_j:=kr+j-1\in[0,K-1]$. *Then* $m_j-1=Kq+u_j$, *and*

$$d(n)=\mathsf{A}_1-K-n+\sum_{j=1}^{k}a_j\,c(m_j),\qquad c(m_j)=\#\{i\ge 1:\ d(i)\le K(q-i)+u_j\}. \tag{4}$$

Proof. (a) The integers in $[1,B_m)$ are the $m-1$ numbers $B_{m'}$ with $m'<m$, together with the $c(m)$ numbers $A_i<B_m$. Hence $B_m-1=(m-1)+c(m)$. Since A is strictly increasing, exactly A_i-i of the B's lie below A_i. So $A_i<B_m$ if and only if $A_i-i\le m-1$, and $A_i-i=Ki+d(i)$.

(b) Because $k\ge 2$ and every $a_j\ge 1$, we have $A_n\ge B_{kn}+B_{kn-1}>B_{kn}$. So at least kn of the B's lie below A_n, and $A_n-n\ge kn\ge m_j>m_j-1$. Since $A_{i+1}-A_i\ge 2$, the quantity A_i-i is strictly increasing in i. So $A_i-i>m_j-1$ for every $i\ge n$, and (a) applies.

(c) By (1) and (a), $A_n=\sum_j a_j(m_j+c(m_j))$, and $\sum_j a_jm_j=K(n-1)+\mathsf{A}_1$. Subtracting $\kappa n=(K+1)n$ gives the first formula. The second is (a) with $m_j-1=Kq+u_j$. As (r,j) ranges over $[0,\tau-1]\times[1,k]$, the number u_j takes each value in $[0,K-1]$ exactly once. □

3.2. **Invariant boxes.** For $0\le r<\tau$ and $x\in\mathbb{Z}$ define

$$\Phi(r,x):=\mathsf{A}_1-K-1-r+\sum_{j=1}^{k}a_j\Big\lfloor\frac{kr+j-1-x}{K}\Big\rfloor.$$

Proposition 3.2 (Box criterion)**.** *Let* $L\le U$ *be integers with*

(i) $\Phi(r,L)\le U$ *for all* r;
(ii) $\Phi(r,U)\ge L$ *for all* r;
(iii) $U\ge -k(\tau-1)$;
(iv) $L\le 0$.

Then $L\le d(n)\le U$ *for all* $n\ge 1$.

Proof. We use strong induction on n, including the base case $n=1$ (where the hypothesis is empty). Let $n=\tau q+r+1$. By Lemma 3.1(b), $c(m_j)=\#\{1\le i<n: Ki\le Kq+u_j-d(i)\}$.

Upper bound. By induction $d(i)\ge L$ for $i<n$, so $c(m_j)\le\#\{i\ge 1: Ki\le Kq+u_j-L\}$. By (iv), $Kq+u_j-L\ge 0$. Hence the count is $\lfloor(Kq+u_j-L)/K\rfloor=q+\lfloor(u_j-L)/K\rfloor$.

Lower bound. By induction $d(i)\le U$ for $i<n$. So every $i\ge 1$ with $i<n$ and $Ki\le Kq+u_j-U$ is counted, i.e. every $i<n$ with $1\le i\le q+\lfloor(u_j-U)/K\rfloor$. We claim that all such i satisfy $i<n$.

- By (iii), $u_j-U\le(K-1)+(K-k)<2K$, so $i\le q+1$.
- If $q\ge 1$, then $q+1\le\tau q<n$.
- If $q=0$ and $r\ge 1$, then $i\le 1<n$.
- If $q=r=0$, then $u_j\le k-1$, and (iii) gives $u_j-U\le K-1$, so no $i\ge 1$ qualifies.

Hence $c(m_j)\ge q+\lfloor(u_j-U)/K\rfloor$.

Inserting both bounds into (4), with $n=\tau q+r+1$ and $\sum_j a_jq=\tau q$, gives $\Phi(r,U)\le d(n)\le\Phi(r,L)$. Then (i) and (ii) close the induction. □

Put $c_0:=\mathsf{A}_1-K-1+(\mathsf{A}_1-\tau)/K$. We have $\sum_j a_j\,kr/K=r$ and $\sum_j a_j(j-1)=\mathsf{A}_1-\tau$. Summing $y-1<\lfloor y\rfloor\le y$ with weights a_j therefore gives

$$c_0-\frac{x}{k}-\tau\ <\ \Phi(r,x)\ \le\ c_0-\frac{x}{k}\qquad(0\le r<\tau,\ x\in\mathbb{Z}). \tag{5}$$

Since $k\ge 2$, the map $x\mapsto c_0-x/k$ is a contraction, so boxes exist. Explicitly:

Proposition 3.3. *Let*

$$L^* = \min\Big\{0,\ \Big\lfloor \tfrac{c_0(1-1/k)-\tau+1}{1-1/k^2} \Big\rfloor,\ \lfloor k(c_0-1)+k^2(\tau-1)\rfloor,\ \Big\lfloor \tfrac{k(c_0-1)}{k+1} \Big\rfloor\Big\}$$

and $U^* = \lfloor c_0 - L^*/k \rfloor$. *Then* (L^*, U^*) *satisfies (i)–(iv) and* $L^* \le U^*$.

Proof. (i) By (5), $\Phi(r, L^*) \le c_0 - L^*/k$, and Φ is an integer, so $\Phi(r, L^*) \le U^*$.

(ii) By (5), $\Phi(r, U^*) > c_0 - U^*/k - \tau \ge c_0(1-1/k) + L^*/k^2 - \tau$. The right-hand side is $\ge L^* - 1$ by the second entry of the minimum. Since Φ is an integer, $\Phi(r, U^*) \ge L^*$.

(iii) We have $U^* > c_0 - L^*/k - 1 \ge -k(\tau-1) - 1$ by the third entry; U^* is an integer, so $U^* \ge -k(\tau-1)$.

$L^* \le U^*$: similarly, $U^* > c_0 - L^*/k - 1 \ge L^* - 1$ by the fourth entry.

(iv) holds because 0 is in the minimum. □

This already proves the weak form of Conjecture 1.1. The following uniform box is sharper and is what makes Theorem B work.

Lemma 3.4 (Universal box)**.** *Let* $L = -K$ *and* $U = \max\{-k(\tau-1),\ \max_r \Phi(r, -K)\}$. *Then (i)–(iv) hold,* $L < U$, *and* $U \le \tau - k$. *Consequently* $-K \le d(n) \le \tau - k$ *for all* $n \ge 1$, *and also for* $n = 0$ *since* $d(0) = 0$ *and* $\tau \ge k$.

Proof. Conditions (i), (iii) and (iv) hold by construction. Also $L = -K < -K + k = -k(\tau-1) \le U$.

$U \le \tau - k$. By (5) and (3), using $(\mathsf{A}_1 - \tau)/K < 1$,

$$\Phi(r, -K) \le c_0 + \tau < \mathsf{A}_1 - K + \tau \le \tau - k + 1.$$

Since Φ is an integer, $\Phi(r, -K) \le \tau - k$. Also $-k(\tau-1) \le 0 \le \tau - k$.

Condition (ii). By (5), $\Phi(r, U) > c_0 - U/k - \tau$. By (3), $c_0 \ge \tau - K - 1$.

- If $U = \max_r \Phi(r, -K)$, then $U \le c_0 + \tau$, and since $1 - 1/k > 0$,

$$\Phi(r,U) > c_0\Big(1-\frac{1}{k}\Big) - \tau\Big(1+\frac{1}{k}\Big) \ge (\tau - K - 1)\Big(1-\frac{1}{k}\Big) - \tau\Big(1+\frac{1}{k}\Big) = -K - 1 + \tau\Big(1-\frac{2}{k}\Big) + \frac{1}{k}.$$

  The right-hand side exceeds $-K - 1$.
- If $U = -k(\tau-1)$, then $\Phi(r, U) > c_0 + (\tau-1) - \tau \ge \tau - K - 2 \ge -K - 1$.

In both cases the integer $\Phi(r, U)$ exceeds $-K - 1$, so $\Phi(r, U) \ge -K = L$. □

Proof of Theorem A. Combine Lemma 3.4 with Proposition 3.2. □

Remark 3.5. (a) *Narrowest boxes.* For fixed $L \le 0$ the least admissible U is $\max\{-k(\tau-1), \max_r \Phi(r, L)\}$, so a narrowest box satisfying (i)–(iv) can be found by a one-dimensional search over L. Every such box contains the range of d. The *exact* range of d can be computed from the automaton of Theorem C (the set of output values of its reachable states); finite prefixes are not reliable for this, since the extreme values can first occur very late. On the 10 named forms of Section 5 and on 736 further forms (all $k = 2$ forms with $a_j \le 20$, all $k = 3$ forms with $a_j \le 6$, and 120 random forms with $4 \le k \le 6$), the narrowest box found by the search equals the exact range of d; an independently written second implementation found the same on 1682 forms. For $(1, 4)$ both are $[-6, 1]$.

Conjecture 3.6. *For every positive integral linear form, the narrowest box satisfying (i)–(iv) is exactly* $[\min d, \max d]$.

(b) *The explicit box.* The box (L^*, U^*) of Proposition 3.3 can exceed the narrowest box by up to 2 units of total width. For example, for $(1, 2)$ it is $[-4, 0]$, whereas the exact range is $[-3, -1]$. In the 736-form sweep the excess is 0, 1, 2 for 49, 277, 410 forms respectively, and never more than 2.

(c) *Sharpness of Lemma 3.4.* Lemma 3.4 bounds the range of d. Its lower end is nearly sharp: in every tested form $\min d \geq -K+2$, and in our 736-form sweep equality holds exactly for the 20 forms $(a,1)$, $1 \leq a \leq 20$ (for example $(1,1)$ and $(2,1)$). We have not proved $d \geq -K+2$ in general. The upper end $\tau - k$ is far from sharp; the second implementation found $\max d \leq 0.66(\tau-k)$ on all its forms.

3.3. **The two-term recursion.**

Proof of Theorem B. Let $n = \tau q + r + 1$ and apply (4). Write $i = q - t$.

- Every $i \leq q-1$ ($t \geq 1$) is counted, because by Lemma 3.4, $d(i) \leq \tau - k < K \leq Kt + u_j$.
- No $i \geq q+2$ is counted, because $K(q-i) + u_j \leq -2K + K - 1 < -K \leq d(i)$.

Hence, for $q \geq 1$,

$$c(m_j) = (q-1) + [\, d(q) \leq u_j \,] + [\, d(q+1) \leq u_j - K \,].$$

For $q = 0$ the count is $c(m_j) = [\, d(1) \leq u_j - K \,]$. This agrees with the same formula, because $d(0) = 0 \leq u_j$. Substituting into (4), with $\sum_j a_j(q-1) = \tau q - \tau$ and $n = \tau q + r + 1$, gives $d(n) = G_r(d(q), d(q+1))$. □

3.4. **Automaticity.**

Proof of Theorem C. For $m \geq 1$ let $V(m) := (d(m-1), d(m), d(m+1)) \in \mathbb{Z}^3$. By Lemma 3.4 (including $d(0) = 0$), $V(m) \in [-K, \tau - k]^3$ for all $m \geq 1$. Fix a digit $0 \leq c < \tau$ and $m \geq 1$. The coordinates of $V(\tau m + c)$ are $d(\tau m + c + e)$ for $e \in \{-1, 0, 1\}$. Write $\tau m + c + e = \tau q' + r' + 1$ with $0 \leq r' < \tau$.

- If $c + e - 1 \geq 0$, then $q' = m$ and $r' = c + e - 1$.
- If $c + e - 1 \in \{-2, -1\}$, then $q' = m - 1$ and $r' = \tau + c + e - 1$, which lies in $[0, \tau - 1]$ because $\tau \geq 2$.

In either case $q' \geq 0$. By Theorem B, each coordinate is $G_{r'}$ applied to two consecutive entries of $V(m)$. So there are maps $H_c : \mathbb{Z}^3 \to \mathbb{Z}^3$, depending only on c and $\mathbf{a}$, with

$$V(\tau m + c) = H_c(V(m)) \qquad (m \geq 1,\ 0 \leq c < \tau). \tag{6}$$

Define a DFAO as follows.

- States: a start state S, plus the vectors $V(m)$ for $m \geq 1$. These lie in $\{S\} \cup [-K, \tau - k]^3$, which has at most $1 + (K + \tau - k + 1)^3$ elements.
- Transitions: $S \xrightarrow{0} S$; $S \xrightarrow{c} V(c)$ for $1 \leq c < \tau$; and $V \xrightarrow{c} H_c(V)$.
- Outputs: S outputs $0 = d(0)$; a vector outputs its middle coordinate.

Feed in the base-τ digits of n, most significant first, with leading zeros allowed. By (6) and induction on the number of digits after the leading zeros, the automaton ends in S if $n = 0$ and in $V(n)$ otherwise. So it outputs $d(n)$. The sequence $(A_n - \kappa n)_{n \geq 1}$ is the restriction of $(d(n))_{n \geq 0}$ to $n \geq 1$, and the same automaton computes it. □

Remark 3.7 (A version with an arbitrary box). The argument does not depend on the specific box of Lemma 3.4. Let $[L, U]$ be any box satisfying (i)–(iv). Let $T = \max(1, \lceil U/K \rceil)$, and let T' be the least integer with $KT' > K - 1 - L$. By (4), for $q \geq T$,

$$d(\tau q + r + 1) = \mathsf{A}_1 - K - 1 - r - \tau T + \sum_j a_j \sum_{t=-T+1}^{T'-1} [\, d(q+t) \leq u_j - Kt \,].$$

Put $o = -2T$, $W = 2T + 2T' + 1$, $P_0 = 2T + 1$, and $V(p) = (d(p+o), \ldots, d(p+o+W-1))$. Then $o \leq 0 \leq o + W - 1$, and the containment conditions

$$\lfloor (o-1)/\tau \rfloor - T + 1 \geq o, \qquad \lfloor (\tau + o + W - 3)/\tau \rfloor + T' - 1 \leq o + W - 1$$

hold for all $\tau \geq 2$. Hence $V(\tau p + c) = H_c(V(p))$ for $p \geq P_0$, with maps H_c on $[L, U]^W$. Iterating, with $b = \sum_{i<e} c_i \tau^i$ (c_0 the least significant digit),

$$V(\tau^e p + b) = H_{c_0} \circ H_{c_1} \circ \cdots \circ H_{c_{e-1}}(V(p)) \qquad (p \geq P_0).$$

So each kernel sequence $p \mapsto d(\tau^e p + b)$ is determined by finitely many data: the map $\pi \circ H_{c_0} \circ \cdots \circ H_{c_{e-1}}$, where π is the coordinate at offset 0, and its first P_0 values. The τ-kernel is therefore finite, and [1, Thm. 6.6.2] applies. With the universal box, $T = 1$ and $T' = 2$.

4. The 4-rusty numbers

Here $\mathbf{a} = (1, 4)$: $k = 2$, $\tau = 5$, $K = 10$, $\kappa = 11$, $\mathsf{A}_1 = 9$. With $u_1 = 2r$, $u_2 = 2r + 1$ and $j = r+1$, Theorem B becomes (2). Conditions (i)–(iv) of Proposition 3.2 hold for $[L, U] = [-6, 1]$. The range of d is exactly $[-6, 1]$ (the output set of the automaton below), so this is the narrowest possible box. Evaluating (2) with $d \in [-6, 1]$ gives, for all $q \geq 0$:

$$\begin{aligned}
d(5q+1) &= -3 + [\,d(q) \leq 0\,], \qquad d(5q+2) = -3,\\
d(5q+3) &= -4 + [\,d(q+1) \leq -6\,] + 4[\,d(q+1) \leq -5\,],\\
d(5q+4) &= -5 + [\,d(q+1) \leq -4\,] + 4[\,d(q+1) \leq -3\,],\\
d(5q+5) &= -6 + [\,d(q+1) \leq -2\,] + 4[\,d(q+1) \leq -1\,].
\end{aligned}$$

In particular $A_{5q+2} = 19 + 55q$ for every $q \geq 0$.

Corollary 4.1. *$A_{5q+1} = 9 + 55q$ if and only if $d(q) \neq 1$. The least q with $d(q) = 1$ is $q = 348$. Hence $A_{5q+1} = 9 + 55q$ for $q < 348$, and $A_{1741} = 19148 \neq 9 + 55 \cdot 348$.*

Proof. $A_{5q+1} = 9 + 55q$ means $d(5q + 1) = -2$, i.e. $d(q) \leq 0$, i.e. $d(q) \neq 1$. From the displayed formulas:

- $d(n) = 1$ if and only if $n = 5q + 3$ with $d(q + 1) = -6$;
- $d(n) = -6$ if and only if $n = 5p$ with $d(p) \geq 0$;
- $d(n) \geq 0$ if and only if either $n = 5q + 4$ with $d(q + 1) \leq -4$, or $n = 5q + 3$ with $d(q + 1) \leq -5$.

Since $d(1), d(2), d(3), d(4) = -2, -3, -4, -5$, the least n with $d(n) \geq 0$ is $5 \cdot 2 + 4 = 14$. Then the least n with $d(n) = -6$ is $5 \cdot 14 = 70$, and the least n with $d(n) = 1$ is $5 \cdot 69 + 3 = 348$. In base 5 the chain is $3 \to 24 \to 240 \to 2343$, and the first failure is at index $5 \cdot 348 + 1 = 1741 = (23431)_5$. The value $A_{1741} = 11 \cdot 1741 - 3 = 19148$ follows from $d(1741) = -3$. □

The automaton of Theorem C for $(1, 4)$ has 28 reachable states, and it minimises to the 17 states of Table 1.

Indexing conventions. Automaton sizes depend on the indexing. Throughout, "minimal" refers to the minimal msd-first DFAO that reads the base-τ expansion of $n \geq 0$, with leading zeros allowed, and outputs $d(n)$, where $d(0) = 0$. Fokkink and Joshi instead feed in $n - 1$, i.e. they describe $s(m) = d(m + 1)$, $m \geq 0$, up to an additive constant [6, Fig. 7]. An automaton for s is obtained from the one of Theorem C by letting S output $d(1)$ and letting each triple output its last coordinate. In that convention the minimal automata are smaller: 9 states for $(1, 4)$, and 6 states for $(1, 3)$, matching the 6-state automaton of [6, Fig. 7]. Table 2 lists both sizes.

Theorem 4.2. *Let $W(n)$ be the output of the DFAO in Table 1 on the base-5 expansion of n. Then $A_n = 11n + W(n)$ is the 4-rusty sequence. This statement is certified by Walnut, independently of Theorems A–C.*

Proof. In Walnut (v7.1) we define the following predicates.

TABLE 1. Minimal msd-first 5-DFAO for $d(n) = A_n - 11n$, $\mathbf{a} = (1, 4)$, in our convention (input $n \geq 0$, leading zeros allowed, state 0 is initial, $d(0) = 0$). This is the Walnut word automaton `RUS`, whose stored outputs are $d + 6$.

state	out	0	1	2	3	4	state	out	0	1	2	3	4
0	0	0	2	4	7	8	9	−5	1	2	3	14	12
1	−1	15	2	4	7	8	10	−6	1	2	4	5	1
2	−2	1	2	4	5	1	11	−6	1	2	4	7	8
3	−3	1	2	4	7	9	12	0	11	2	4	7	8
4	−3	1	2	4	6	12	13	0	11	2	4	7	9
5	−4	1	2	4	7	8	14	1	10	16	4	7	9
6	−4	1	2	4	7	9	15	−2	1	2	4	7	8
7	−4	1	2	3	13	12	16	−3	1	2	4	5	1
8	−5	1	2	4	7	8							

- ra(n, s): $n > 0 \wedge s + 6 = 11n + \texttt{RUS}[n]$.
- rb2(n, t): $\exists s$ ra$(n, s) \wedge (5t = s + 1 \vee 5t = s + 2)$.
- rb1(n, u): $\exists s, t$ ra$(n, s) \wedge$ rb2$(n, t) \wedge u + 4t = s$.

Walnut returns `TRUE` for each of the following.

- Initial values: rb1$(1, 1)$, rb2$(1, 2)$ and ra$(1, 9)$.
- Totality: ra and rb2 are defined for all $n > 0$.
- Monotonicity: ra is strictly increasing, and $B_1(n) < B_2(n) < B_1(n + 1)$.
- Disjointness: the range of ra meets neither range of rb1, rb2.
- Cover: every $x > 0$ lies in one of the three ranges.
- Bounds: $11n - 6 \leq A_n \leq 11n + 1$.
- First failure: the first failure of $A_{5n+1} = 9 + 55n$ is at $n = 348$.

Control queries, e.g. ra$(1, 10)$, and a cover statement omitting ra, return `FALSE`. The relation $A_n = B_1(n) + 4B_2(n)$ holds by construction of rb1. The interleaving makes (B_m), defined by $B_{2n-1} = B_1(n)$ and $B_{2n} = B_2(n)$, strictly increasing. So (A, B) is a complementary pair of strictly increasing sequences satisfying (1), and by uniqueness [5, Lemma 3], [6, Lemma 2] it is the anti-recurrence pair of $(1, 4)$. A second, independently written query set gives the same verdicts (artifact files `walnut_ref14.*`). □

5. Computational verification

None of the following is needed for the proofs. Its purpose is to rule out slips in the hand arguments. All computations ran on a single CPU core under a 500 MB memory cap. Every check generates (A, B) by brute force directly from the definition (mex plus linear form), independently of the formulas being tested.

(1) *Large-scale check for* $(1, 4)$.
   - 10^7 terms: $d \in [-6, 1]$, and (2) holds for every $n \leq 10^7$.
   - The 17-state DFAO agrees with $d(n)$ for every $n \leq 2 \cdot 10^6$.

(2) *Step-by-step recheck on* 50 *forms (*20 *not* A_1*-bounded),* $6 \cdot 10^4$ *terms each.* Checked:
   - $B_m = m + c(m)$ and identity (4);
   - conditions (i)–(iv) for the explicit box, and the pointwise bounds $\Phi(r, U) \leq d(n) \leq \Phi(r, L)$;
   - the window recursion.

   No failures.

(3) *An independently written second implementation (*585 *forms).* This and item 5 were written separately from our code, as a check against implementation errors; they are not peer review.
- Forms: all $k = 2$ with $a_j \le 12$; all $k = 3$ with $a_j \le 6$; 25 adversarial forms; 200 random forms with $k \in \{3, 4, 5\}$ and $a_j \le 12$. Of these, 178 are not A_1-bounded.
- Up to 10^5 terms per form, and $2 \cdot 10^6$ terms for 8 forms.
- Code: a C generator written separately from our code.
- Checked: the identity, the box, the window recursion, and determinism of the window map. No failures.

(4) *Checks of the present formulation.*
- 433 forms, 6000 terms each: Lemma 3.4, the bound $U \le \tau - k$, and Theorem B for all $q \ge 0$ with $d(0) = 0$. Forms: all $k = 2$ with $a_j \le 12$; all $k = 3$ with $a_j \le 5$; 150 random forms with $k \in \{3, 4, 5\}$ and $a_j \le 12$; extremal forms $(1, \dots, 1)$, $(1, \dots, 1, 20)$, $(20, 1, \dots, 1)$. No failures.
- 17 forms, $2 \cdot 10^4$ terms: the explicit constants of Proposition 3.3 and Remark 3.7, verified coordinate by coordinate.
- 12 named forms: the automaton of Theorem C reproduces $d(0), \dots, d(2{\cdot}10^4)$ exactly. After minimisation it has 17, 19, 37 and 24 states for $(1, 4)$, $(1, 5)$, $(2, 7)$ and $(1, 1, 5)$ respectively. These numbers match the Walnut-certified automata of Section 6.
- 10 named forms and 736 further forms (Remark 3.5): exact ranges from the reachable states of the automaton of Theorem C, narrowest boxes, the excess of (L^*, U^*), $V(1) \in [-K, \tau - k]^3$, and minimal sizes in both conventions.

(5) *A second pass of the separately written implementation (*2270 *forms).*
- Forms with $2 \le k \le 8$, entries up to 200 and τ up to 1600.
- Checked: Lemma 3.4, Theorem B for all $q \ge 0$, the maps H_c of (6), and that the reachable triples of Theorem C lie in $[-K, \tau - k]^3$. No failures.
- $3 \cdot 10^6$ terms for 7 large forms; for $(1, 4)$, $A_{1741} = 19148$ and $A_{5q+2} = 19 + 55q$ for all $5q + 2 \le 2 \cdot 10^6$.
- Exact rational recheck of the constants of Proposition 3.3 and Remark 3.7 on 1420 forms.

6. Further Walnut certificates

For the forms in Table 2, a DFAO was built from the window recursion and minimised. Its outputs were stored as a Walnut word automaton, and the analogue of the query set in Theorem 4.2 was run:

- initial block, totality, monotonicity of A, interleaving of the B's;
- disjointness of A from the B's, and cover of $\mathbb{N}$;
- the linear relation, plus control queries.

All verdicts were as expected. The B's are pairwise disjoint because of the interleaving query $B_1(n) < \cdots < B_k(n) < B_1(n + 1)$. For $(1, 1, 5)$ the separately written implementation also ran the three pairwise disjointness queries for B_1, B_2, B_3 directly, with its own predicate definitions; all return `TRUE` (artifact files `walnut_rqf_disjB.*`). None of these forms is A_1-bounded, so none is covered by [5, 6].

For $(1, 1, 5)$ the direct query for the linear relation exceeded a 400 MB heap. It was replaced by an equivalent reformulation that first defines the sum $B_1 + B_2 + 5B_3$ as a predicate; both resulting queries return `TRUE`. For $(1, 5)$, $(2, 7)$ and $(1, 1, 5)$ the relation is checked by an explicit Walnut query, not just by construction. These certificates are not needed given Theorem C; they serve as an independent check of it.

TABLE 2. Walnut-certified forms (Walnut v7.1). "States" is the size of the minimal msd-first DFAO in our convention (input $n \geq 0$, output $d(n)$, $d(0) = 0$); "FJ" is the size in the convention of [6] (input $n-1$). The range of d is exact.

a	τ	κ	states	FJ	range of d	Walnut automaton
$(1,4)$	5	11	17	9	$[-6,1]$	`RUS`
$(1,5)$	6	13	19	11	$[-7,1]$	`RFV`
$(2,7)$	9	19	37	20	$[-12,3]$	`RTS`
$(1,1,5)$	7	22	24	9	$[-9,-1]$	`RQF`

7. Related work

Prior art search. We searched for any earlier proof of boundedness or automaticity for general linear forms (arXiv, zbMATH Open, OEIS, publisher pages; October 9, 2026). Except where noted, we read each source listed here.

- *Kimberling–Moses* [12]. Theorem 2.1 and Theorem 3.2 determine growth rates for linear complementary equations. Section 6 introduces the anti-Fibonacci system and its anti-m-nacci generalisations, with conjectured morphisms for the difference sequences, and asks whether the anti-m-nacci sequence is the fixed point of a morphism ("The conjectured answer is yes!"). Example 12 conjectures bounded discrepancy for a related system. No general boundedness or automaticity theorem is stated.
- *Kimberling–Moses 2021* [13]. Theorem 1 gives existence and uniqueness for complementary equations $a_n = f(b_{g(n)}, b_{h(n)})$, including $a_n = b_{2n} + b_{2n+1}$; explicit results are proved for $a_n = b_{2n} + b_{4n}$. Section 6 studies $a_n = ub_n + vb_{n+k} + c$, where tables and experiments are said to suggest that $a_n - (u+v+1)n$ is unbounded unless $u = v = 1$. There is no general boundedness result for the block equations (1).
- *Bosma et al.* [3]. Section 3 proves the anti-tribonacci and anti-tetranacci conjectures of Kimberling by Walnut verification of guessed automata, and states the Clergyman's Conjecture for anti-k-naccis.
- *Fokkink–Joshi* [5, 6]. Both versions prove the A_1-bounded case and the anti-k-bonacci case, and state that the general case is open [6, §6]. The published version was received 8/6/25 and published 1/19/26.
- *Citation and phrase searches.* We found no paper citing [5, 6] that treats anti-recurrences. Searches of arXiv and zbMATH Open for "anti-recurrence", "anti-Fibonacci", "anti-k-nacci", "anti-bonacci", and "Clergyman" found nothing on anti-recurrences beyond the works cited here.
- *OEIS* [16, 17, 18, 19, 20]. The bounds in these entries are stated as conjectures, or have been proved for those specific forms only (by [3, 5]). The 3- and 4-rusty sequences are not in the OEIS.
- *Older complementary-equation surveys* [10, 11]. These treat equation families such as $b(n) = a(jn) \pm r$, $b(n) = a(jn) + kn$ and Wythoff-type equations, not mex-generated linear forms.
- *Walnut and complementarity.* Shallit [21] proved complementarity statements for Tribonacci-based sequences in Walnut; the setting is different. For Walnut in general see [22, 15].
- *Mex-defined complementary pairs.* Fraenkel [7], which [6] cites as an origin of complementary sequences, studies P-positions of generalised Wythoff games (we have seen only its opening section). Fraenkel and Peled [8] study pairs $a_n = \mathrm{mex}\{\dots\}$, $b_n = a_n + c_n$ and

show (zbMATH summary; full text not seen) that membership is decidable in polynomial time when c_n is approximately linear. Neither treats (1).

We could not run Google Scholar or MathSciNet searches. To the best of our knowledge, boundedness (the weak form) and automaticity for general linear forms are new. So are the explicit universal box, the two-term recursion, and the explanation of the A_{1741} break. [TODO: repeat the search on Google Scholar and MathSciNet, and read Fraenkel 1982 and Fraenkel–Peled in full, immediately before submission.]

Background. Counting arguments of the kind in Lemma 3.1(a) go back to Lambek and Moser's work on complementary sequences [14]. Automatic sequences go back to Cobham [4]; see [1] for the theory. Since automatic sequences and linear sequences are τ-regular and τ-regular sequences form a ring [2], Theorem C also shows that A_n itself is τ-regular.

Relation to the A_1-bounded proof. Fokkink and Joshi show that under A_1-boundedness each interval $I_n = [\kappa(n-1)+1, \kappa n]$ contains exactly one A-term. The process is then a uniform substitution on residues, which they handle with Cobham's little theorem. Without A_1-boundedness this interval structure breaks down; for example the 6-state automaton for $(1,3)$ [6, Fig. 7] has all 0-transitions returning to the initial state, a feature that fails for $(1,4)$ [6, §5]. For $(1,4)$, d takes the value 1, so A_n can leave I_n. Our counting identity avoids intervals altogether: it reads $d(\tau q + r + 1)$ off the two values $d(q)$ and $d(q+1)$.

8. Data and code availability

All scripts, logs and Walnut files are in the reproducibility artifact accompanying this paper (directory `artifact/`; see its `README.md` and `SHA256SUMS`). It contains:

- Python generators and checkers;
- two separately written C generators with their checkers (in `code/referee/` and `code/referee2/`);
- the Walnut word automata `RUS`, `RFV`, `RTS`, `RQF`;
- the Walnut command files and their output logs;
- the scripts that build the automaton of Theorem C, and the exact-range and automaton-size computations (`r2checks.py`).

Walnut is free software [15]; we used version 7.1. [TODO: deposit the artifact (e.g. Zenodo) and insert DOI.]

Use of AI tools

[TODO: Draft disclosure; the author must edit and complete it before submission.] This work was carried out almost entirely by AI agents working under the author's direction. AI agents found and wrote the proofs in Section 3 and Section 4, wrote all code (the generators, the checkers, the automaton constructions and the Walnut query files), ran all computations and Walnut verifications, carried out the literature search, and drafted this paper. The two internal review passes and the separately written second implementation mentioned in Section 5 were also carried out by AI agents; they are not peer review. The Walnut certificates of Theorem 4.2 and Section 6 were produced by Walnut itself and can be re-run from the artifact. [TODO: State precisely what the author personally checked, e.g.: "The author has read and verified the proofs of Lemma 3.1, Proposition 3.2, Lemma 3.4 and Theorems A–C, re-ran the Walnut files …, and checked the citations …" Only list what was actually done.] The author takes full responsibility for the content.

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